DRUGS · THYROID AND PARATHYROID

Active vitamin D: alfacalcidol and calcitriol

Alfacalcidol and calcitriol are active forms of vitamin D that do not need renal 1-alpha-hydroxylation, which depends on parathyroid hormone. They are therefore the right vitamin D for patients with hypoparathyroidism after thyroid or parathyroid surgery, and for hungry bone syndrome. Ordinary vitamin D (colecalciferol) is used to correct deficiency, including before parathyroidectomy.

1 CHAPTER ONE At a glance

The drugs on one page.

1.1 Summary

ACTIVE VITAMIN D

	SUMMARY
Alfacalcidol	1-alpha-hydroxycholecalciferol, converted in the liver to calcitriol; adult 0.5 to 1 microgram daily initially, adjusted to calcium (commonly 0.25 to 2 micrograms daily); child by weight
Calcitriol	1,25-dihydroxycholecalciferol (active hormone); 0.25 to 0.5 micrograms once or twice daily
Colecalciferol	Native vitamin D3 for deficiency (for example a loading course then maintenance by local protocol)
How they work	Increase intestinal absorption of calcium and phosphate and bone resorption; raise serum calcium within days
ENT uses	Hypoparathyroidism after thyroidectomy (with oral calcium); hungry bone syndrome after parathyroidectomy (often started before surgery in severe disease); vitamin D deficiency before parathyroidectomy
Monitoring	Calcium two or three times a week at first, then weekly to monthly; aim for low normal calcium in hypoparathyroidism to limit hypercalciuria

2 CHAPTER TWO Using them in ENT

Practice.

2.1 After thyroidectomy

When PTH is low after total thyroidectomy, alfacalcidol is often started with oral calcium; wean both as parathyroid function recovers, guided by calcium and PTH, usually over weeks.

2.2 Before parathyroidectomy

Correct vitamin D deficiency before surgery for primary hyperparathyroidism (it does not usually worsen hypercalcaemia and reduces post-operative hungry bone effects).

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Hypercalcaemia (thirst, polyuria, constipation, confusion) and hypercalciuria with nephrocalcinosis and stones if overtreated; alfacalcidol and calcitriol have a short half-life, so hypercalcaemia resolves within days of stopping.

3.2 Interactions

Thiazides (hypercalcaemia); digoxin; enzyme inducers reduce vitamin D levels.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Why **active** vitamin D is needed in hypoparathyroidism.
- Calcium targets in long-term hypoparathyroidism.

4.2 Common pitfalls

COMMON PITFALLS

- Using colecalciferol alone in post-surgical hypoparathyroidism.
- No calcium monitoring after starting alfacalcidol.

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